

MSc in Computer Science and Engineering

Course: Software Project Management and Quality Assurance

Class Assignment

Topic: Development Planning, Quality Planning and Software Risk Management Case Study

The Department of Computer Science and Engineering (CSE), Jahangirnagar University intends to replace its existing student information management process with a web-based **Student Information System (JU CSE SIS)**.

The proposed system will support:

- Student admission and profile management
- Course registration
- Semester-wise result processing
- Attendance management
- Advisor approval
- Tuition fee tracking
- Class routine generation
- Notice and announcement management
- Faculty information management
- Transcript generation
- Integration with university authentication services
- Future integration with university finance and library systems

The project duration is **10 months** with a budget of **BDT 40 lakh**. The development team consists of:

- 1 Project Manager
- 1 Business Analyst
- 4 Software Developers
- 2 QA Engineers
- 1 Database Administrator
- 1 DevOps Engineer

The department expects the system to be deployed before the next academic semester.

Assignment Questions

Question 1 (20 Marks)

Development Plan

Prepare a **Development Plan** for the proposed JU CSE SIS.

Your plan should include appropriate information for the following components:

1. Project Deliverables
2. Project Interfaces
3. Development Methodology and Development Tools
4. Software Development Standards and Procedures
5. Development Process Map
6. Project Milestones
7. Team Organization and Responsibilities
8. Required Development Facilities
9. Development Risks and Risk Management Activities
10. Project Control Methods
11. Project Cost Estimation

Students are expected to justify every decision instead of merely listing items.

Learning Outcome: Identify and apply the elements of a Development Plan.

Question 2 (20 Marks)

Software Quality Plan

Prepare a **Software Quality Plan** for JU CSE SIS.

Your report should include:

- Quality Goals (with measurable metrics)
- Planned Review Activities
- Software Testing Strategy
- Acceptance Testing Plan
- Configuration Management Plan
- Change Management Process

For each quality activity, explain:

- objective,

- responsible person,
- timing,
- expected deliverables.

Use measurable quality metrics whenever possible.

Learning Outcome: Identify and apply the elements of a Software Quality Plan.

Question 3 (20 Marks)

Software Risk Analysis

Identify **at least eight software risks** that may occur during the development of JU CSE SIS.

For each risk, provide:

- Risk description
- Risk category
- Probability (Low/Medium/High)
- Impact (Low/Medium/High)
- Mitigation strategy
- Contingency plan

Include risks from different categories such as:

- Scheduling
- Requirements
- Personnel
- Technology
- Resource
- Subcontracting (if applicable)

Learning Outcome: Identify the major software risk items and understand software risk management.

Question 4 (15 Marks)

Risk Prioritization

The following events occur during the project:

- Two senior developers resign.
- University changes grading policy.
- LDAP authentication API is delayed.

- Server procurement is postponed.
- Mid-semester examination schedule changes.

Rank these risks from highest to lowest priority.

Justify your ranking using:

- Probability
- Impact
- Risk Exposure

Question 5 (15 Marks)

Quality Improvement

Suppose that during System Testing:

- 180 defects are reported.
- 35 defects are Critical.
- Response time exceeds 5 seconds during course registration.
- User Acceptance Testing reports poor usability.

As the Quality Manager:

Explain

1. Which quality goals have failed?
2. Which review activities should have detected these earlier?
3. What corrective actions should be taken?
4. Which metrics should be monitored before release?

Question 6 (10 Marks)

Software Risk Management Process

Using the software risk management process discussed in the lecture, explain how the JU CSE SIS project team should manage software risks throughout the project lifecycle.

Your discussion should cover:

- Risk Identification
- Risk Assessment
- Risk Planning
- Risk Mitigation
- Monitoring

- Risk Review

Illustrate the process using examples from the JU CSE SIS project.

Conceptual Questions (Understanding Test)

Answer the following briefly.

1. Why is a Development Plan insufficient without a Quality Plan?
2. Explain the difference between project milestones and project deliverables.
3. Why should quality goals be measurable rather than qualitative?
4. Explain the difference between risk mitigation and contingency planning.
5. Why is configuration management considered an essential part of a Software Quality Plan?
6. Which development plan components are most affected when project requirements change?
7. Explain how poor requirement management can increase project risk.
8. Why should acceptance testing criteria be defined before software development begins?
9. Explain the relationship between project control methods and software quality.
10. Which software risks are most critical for a university-wide Student Information System and why?

Higher-Order (Critical Thinking) Questions

1. Suppose JU decides to expand the SIS for all faculties after six months of development. How should the Development Plan and Quality Plan be revised?
2. Compare Agile Scrum and Waterfall for developing JU CSE SIS. Which would you recommend? Justify your answer considering quality assurance and risk management.
3. If the project budget is reduced by 25% while the deadline remains unchanged, which Development Plan elements should be revised first? Explain your reasoning.
4. Propose three quantitative quality metrics that should be monitored during development and explain how they support project success.
5. Based on Boehm and Ross's Top 10 Software Risk Items, identify the three most significant risks for JU CSE SIS and justify why they pose the greatest threat to project success.